```
   gets(pat);
   printf("\n enter replace string");
   gets(rep);
   stringmatch();
   if(flag==1)
      printf("\n resultant string is %s",ans);
   else
      printf("\n pattern string not found");
   return 0;
}
```

```
enter main string WELCOME TO AIDS

enter pat string AIDS

enter replace string CSBS
```

3. stack

3.Stack

```
#include <stdio.h>
#include <stdlib.h>
#include <conio.h>
#define MAX 4
int stack[MAX],item;
int ch,top=-1,count=0,status=0;
void push()
{
int item;
if(top ==(MAX-1))
printf("\n\n stack is overflow");
else
{
   printf("enter element to be pushed");
   scanf("%d",&item);
stack[++top] = item;
}
}
void pop()
{
```

```
if(top== -1)
printf("\n\n stack is underflow");
else
{
printf("\n popped element is %d",stack[top--]);
}
}
void display()
{
int i;
if(top==-1)
printf("the stack is empty");
else
{
 printf("\n stack contents are:");
for(i=top;i>=0;i--)
printf("stack[%d]=%d\n",i,stack[i]);
printf("\n");
}
}
void palindrome()
{
   int i,count=0;
   for(i=0;i<=(top/2);i++)
   {
      if(stack[i]==stack[top-i])
         count++;
   }
   if((top/2+1)==count)
      printf("stack contents are palindrome");
   else
      printf("stack contents are not palindrome");
}
int main()
{
int choice;
while(1)
{
```

```
printf("\n1.push\n 2.pop\n 3.Display\n 4.Palindrome\n  5.Exit\n");
printf("\nEnter your choice: ");
scanf("%d",&choice);
switch(choice)
{
case 1:push();
break;
case 2:pop();
break;
case 3:display();
break;
case 4:palindrome();
break;
case 5:return 0;
default: printf("end of execution\n");
}
}
```

5. Develop a Program in C for the following Stack Applications

a. Evaluation of Suffix expression with single digit operands and operators: +, -, *, /, %, ^

b. Solving Tower of Hanoi problem with n disks.

```
#include<stdio.h>
#include<conio.h>
#include<math.h>
#include<string.h>
double compute(char symbol, double op1, double op2)
{
switch(symbol)
{
case '+': return op1+op2;
case '-': return op1-op2;
case '*': return op1*op2;
case '/':return op1 / op2;
case '$':
case '^': return pow(op1,op2);
default: return 0;
}
```